

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

There were 12 files, 1 file for each month. All the data files were kept on google drive and then the google drive was mounted inside the code to avoid uploading the files again and again. All the warnings and needed libraries were successfully loaded needed for sampling.

Approach: The sampled file approximately contained 1.9 million records so we can assume the total records were somewhere around 38 million as we have sampled 5 percent of the data. Instead of gathering data by a random selection, the dataset was sampled by selecting 5% of trips for each hour of each day across all monthly files by looping over each file and then randomly selecting data for each hour and capturing 5 percent of the data. Then data for each hour for each day was being appended into a single sampled file.

This approach ensured balanced representation of data during peak and off-peak hours, avoids any biases, and preserves hourly and daily demand structure and reducing computational load.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

Removed new/extra columns date and hour were generated as a part of sampling the data by each day and each hour. Also after concatenation operations and combining all the data together the index was reset to maintain sequential indexing on the newly generated file and avoided duplicates and messy indices. This was done using pandas data frame `reset_index` inbuilt feature.

Negative values like `fare_amount`, `extra`, `Airport_fee`, `surcharge` columns represent illogical and invalid values. These rows were also removed to ensure that all monetary fields contain valid non-negative values

2.1.2. Combine the two airport_fee columns

Two airport fee columns (`Airport_fee` and `airport_fee`) were merged by selecting the non-null value for each record using horizontal backfilling. Since the values were mutually exclusive, they were merged by retaining

the non-null value for each record and the redundant column was dropped. This avoided duplicate columns and confusion for future references to Airport_fee column.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Used Mean to calculate the proportion of missing values in each column by counting NaN values and dividing them by the total number of rows. Then columns were sorted in descending order to prioritise variables with more NaN values.

The columns observed were:

Airport_fee	3.42
congestion_surcharge	3.42
passenger_count	3.42
RatecodeID	3.42
store_and_fwd_flag	3.42

Rest all other columns had 0 missing value proportion.

2.2.2. Handling missing values in passenger_count

There were around 64874 rows having passenger count as null and 29681 values having passenger count as 0. Used median to handle missing values in passenger_count because it is a discrete variable and is good when we have outliers so rows containing missing values were identified using a null check and then values were imputed using median to maintain data consistency. Median calculated= 1. Post cleaning there were no null or zero values in passenger_count column

2.2.3. Handle missing values in RatecodeID

There were 64874 rows having RatecodeID as NaN. These missing values in the RatecodeID column were imputed using the mode, as it is a categorical variable and the most frequent category best represents the missing entries. Mode calculated= 1. Post cleaning there were no null or zero values in RatecodeID column

2.2.4.Impute NaN in congestion_surcharge

There were 64874 rows having congestion_surcharge as NaN. Missing values in the congestion_surcharge column were replaced with 0, as NaN values of surcharge logically indicates that no congestion fee was applied. This ensures accurate revenue and surcharge frequency calculations. Post cleaning there were no null values in congestion_surcharge column
Other columns were also handled with any missing values using the same approach.

2.3. Handling Outliers and Standardising Values

2.3.1.Check outliers in payment type, trip distance and tip amount columns

Firstly trips with passenger_count greater than 6 were removed as these values look unrealistic for standard taxi rides and can be concluded as outliers.

Removed erroneous values:

1. A trip with zero or close to zero distance cannot cost fare amount greater than 300.
2. A trip having distance as 0 and fare amount = 0 but pickup zone is different from dropoff zone is again an erroneous entry
3. A trip having distance of more than 250 miles can be considered as an outlier for a NYC taxi ride
4. Payment type cannot be 0

Above outliers were handled. Records having logically incorrect or unrealistic values were removed, while valid columns were standardised to ensure data integrity and cleanliness before further analysis.

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1.Classify variables into categorical and numerical

Categorise the variables into Numerical or Categorical.

VendorID: Categorical

tpep_pickup_datetime: Temporal (Neither Numerical nor Categorical)

tpep_dropoff_datetime: Temporal (Neither Numerical nor Categorical)

passenger_count: Numerical

trip_distance: Numerical

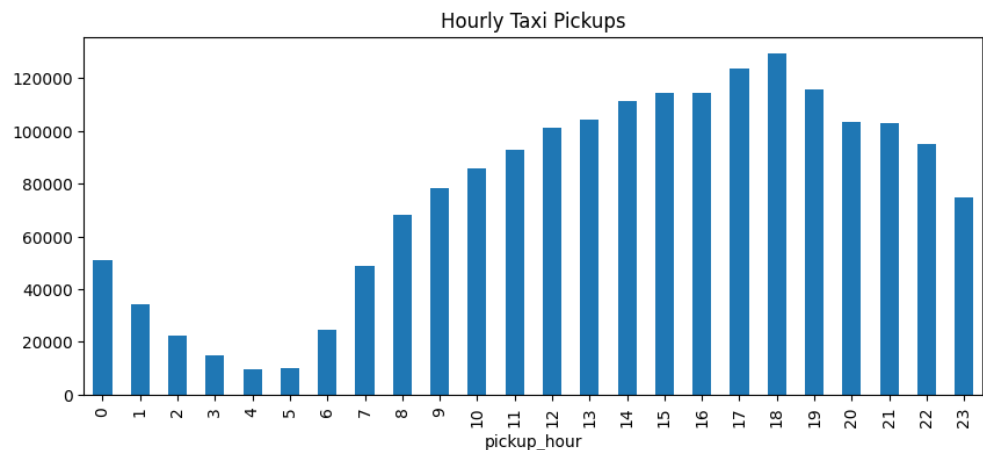
RatecodeID: Categorical

PULocationID: Categorical

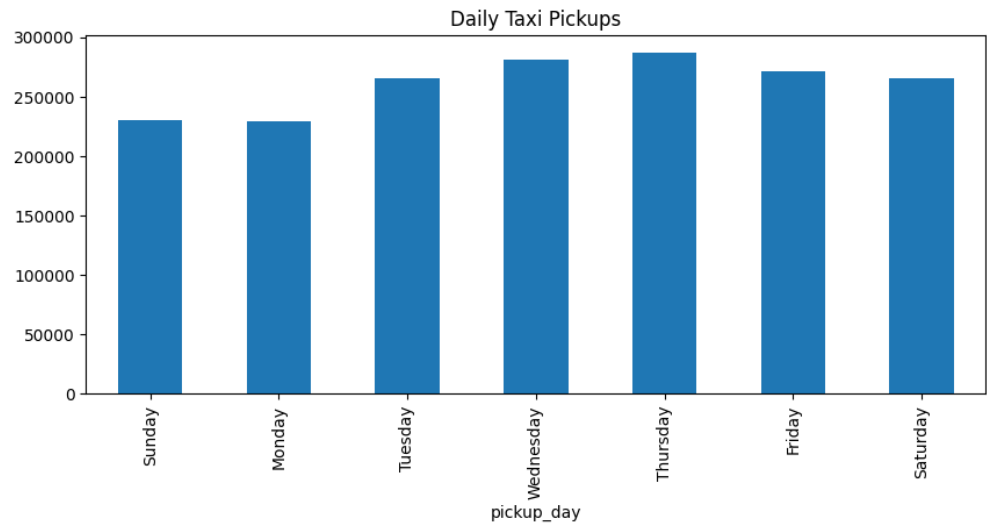
DOLocationID: Categorical

payment_type: Categorical
pickup_hour: Numerical
trip_duration: Numerical
fare_amount: Numerical
extra: Numerical
mta_tax: Numerical
tip_amount: Numerical
tolls_amount: Numerical
improvement_surcharge: Numerical
total_amount: Numerical
congestion_surcharge: Numerical
airport_fee: Numerical

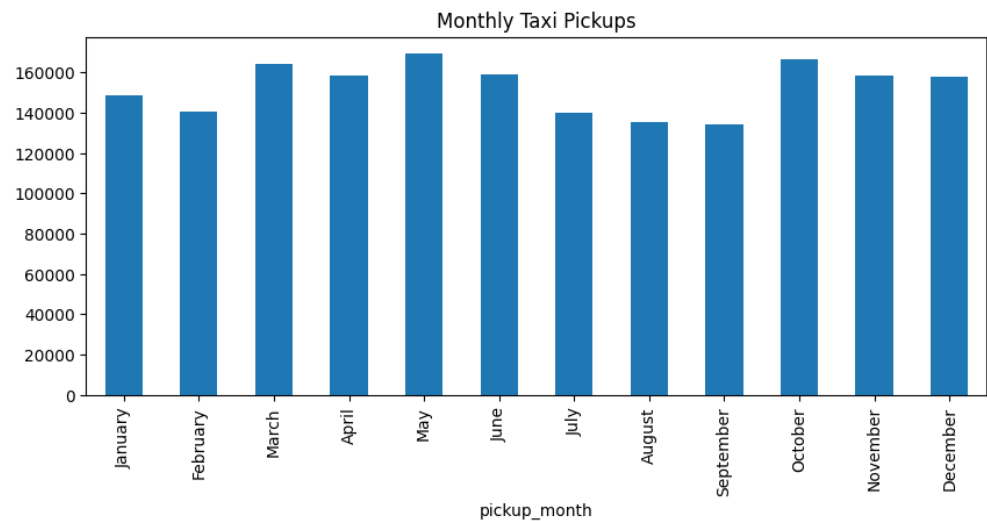
3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months



Hourly trends in taxi pickup: The hourly pickup distribution tells us that rides are peaking during morning commute hours 7-9 AM (We can assume office goers at this point of time) and attain a maximum busiest peak in the evening between 4-7 PM (we can assume people returning from offices and some people heading for food/dining/leisure), with the highest pickups recorded around 6 PM. At night there's a gradual decline as bookings are relatively low as compared to the day. Very few rides are booked between 3-5 AM, indicating minimal travel activity during early morning hours. We can also tell that maximum number of fleet is operating during the day time.



Daily trends in taxi pickup: The daily pickup distribution indicates that demand peaks during mid-week, particularly on Wednesday and Thursday. On the Weekend, the demand is significant but still lower than peak weekday demand having Sunday as the lowest pickups. Here we can assume there is more work-related travel in shaping overall taxi demand.



Monthly trends in taxi pickup: The monthly distribution of taxi pickups shows a moderate demand throughout. Demand increases steadily from March to May, with May recording the highest pickups. A noticeable decline is seen during July-September, which could be possibly due to vacation season and less business travel. Demand rises again in October and a similar demand is seen in November and December which could possible represent tourism and holiday activity.

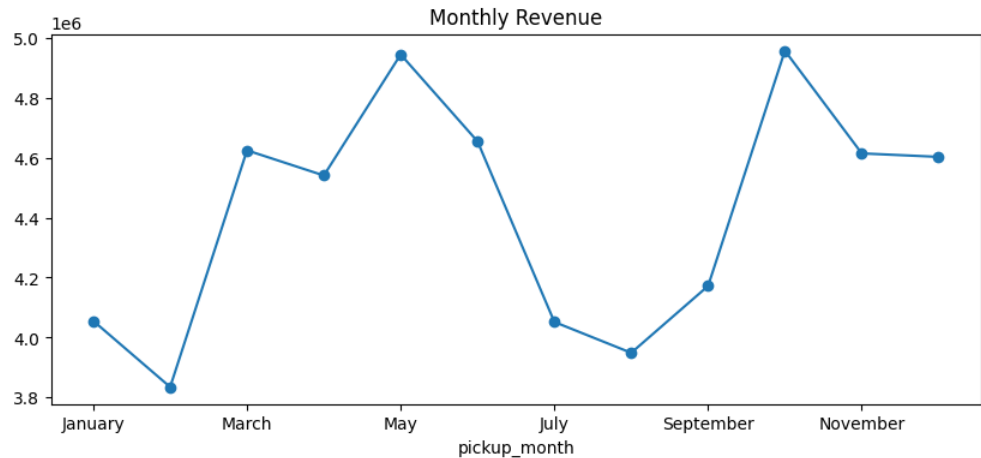
3.1.3. Filter out the zero/negative values in fares, distance and tips

While cleaning the data in the start, we had already removed any negative values. So no negative values are there for any of the columns.

For zero values: No, removing all zero values would not be right, as in the case of tip_amount zero values can represent real world scenarios and also trips within the same pickup and drop-off zone, so it may create a some change in understanding data analysis.

Remove fare amount and total_amount as zero values is an invalid scenario however removing tip amount and trip distance would not be right as they seem valid values.

3.1.4. Analyse the monthly revenue trends



Monthly revenue follows a similar pattern as we saw in the monthly pickup analysis. Revenue is maximum in May and October, indicating high demand during spring and fall seasons. A noticeable dip occurs during July and August, which is likely due to reduced business and leisure activity during summer months. February records the lowest revenue, possibly influenced by weather and shorter month duration. Overall, revenue variations are primarily demand-driven rather than pricing-drive.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

total_amount	
quarter	
1	23.61
2	26.68
3	22.97
4	26.74

Quarterly revenue analysis shows clear seasonal variation. Q2 and Q4 contribute the highest revenue shares (approximately 26.7% each), indicating strong demand during spring and late-year periods. Q3 contributes the lowest share (22.97%), reflecting reduced activity during summer months. This seasonal pattern highlights the importance of adjusting fleet allocation and pricing strategies across quarters to maximise revenue.

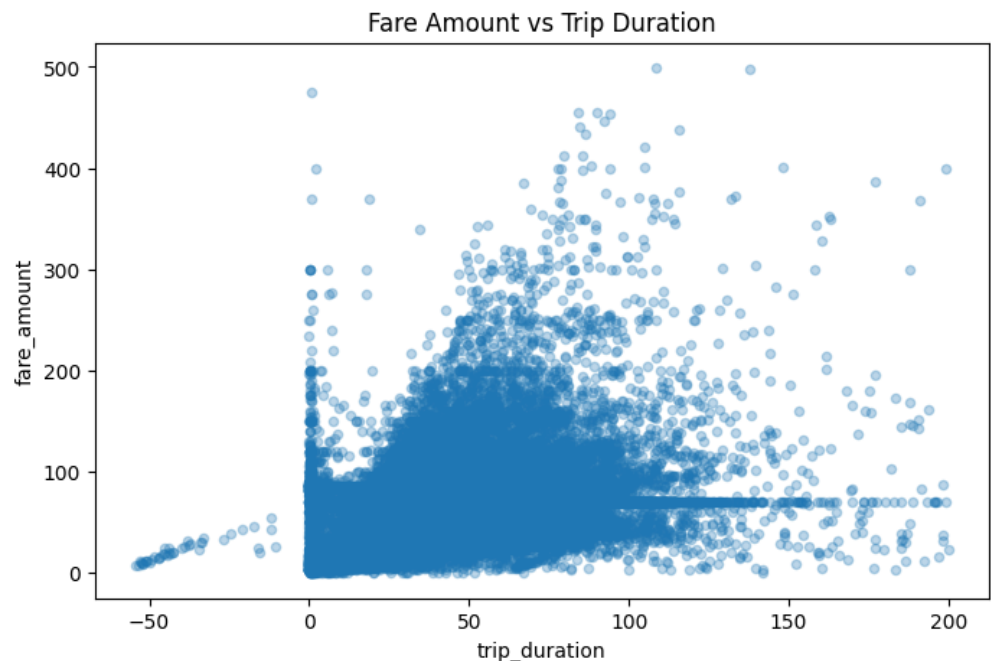
3.1.6. Analyse and visualise the relationship between distance and fare amount



Trip Distance vs Fare amount:

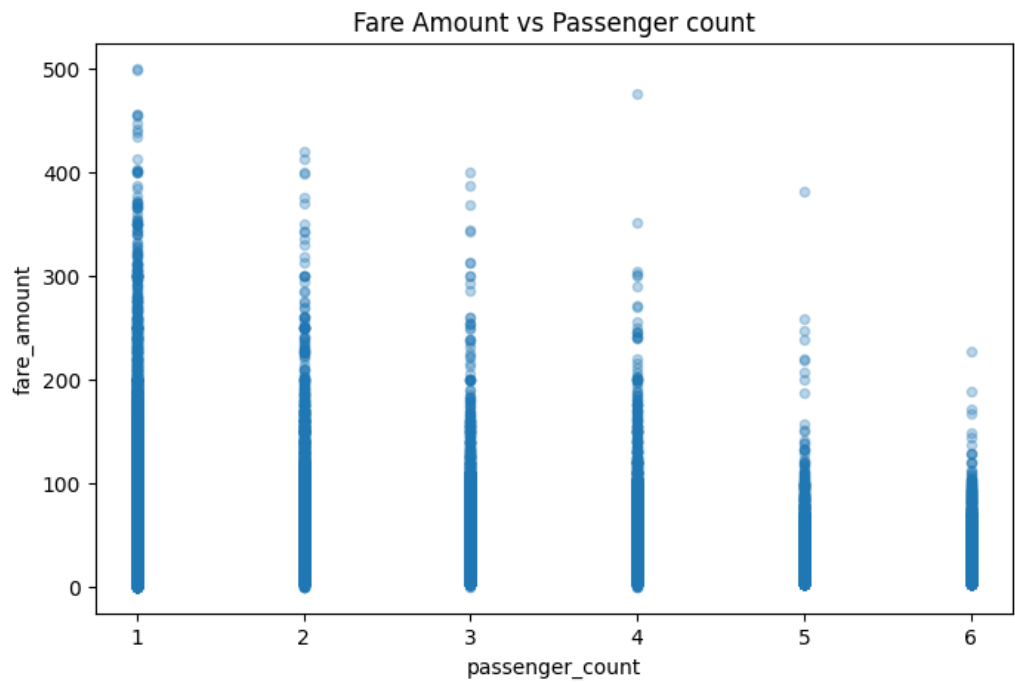
1. There is a very strong positive relationship between distance and fare.
2. As trip distance increases, the fare amount also increases.
3. Short trips (0–10 miles) are very common. Very long trips are rare but have much higher fares.
4. The spread becomes slightly wider for longer distances, meaning some longer trips vary in price due to other factors.
5. Very strong correlation between Trip Distance and fare amount, depicting distance based pricing

3.1.7. Analyse the relationship between fare/tips and trips/passengers



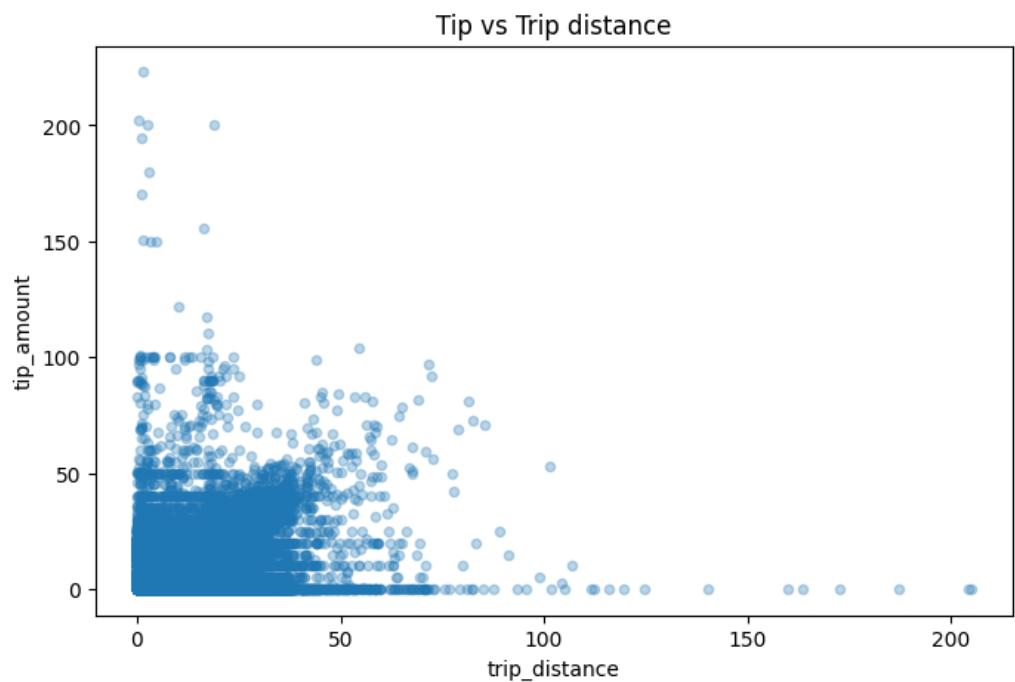
Fare amount vs trip duration:

1. There is an upward trend as trip duration increases, the fare amount generally increases.
2. Most trips fall between 0 to 100 minutes.
3. There is wider variation in fares for longer durations.
4. Some trips with long durations do not always have very high fares, likely due to traffic conditions or route variations.
5. A few extreme values are visible but do not dominate the overall pattern.
6. Strong correlation depicting fare is influenced not only by distance but also by time.



Fare Amount vs Passenger count:

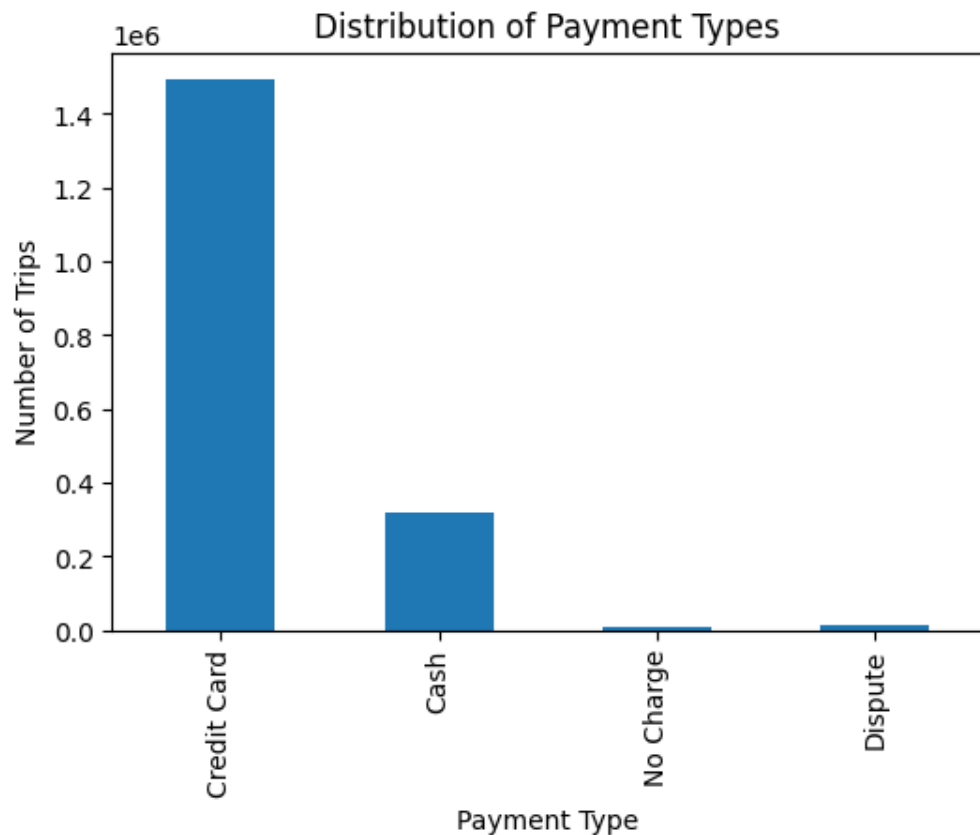
1. There is no clear upward or downward trend.
2. Fare amounts are spread similarly across all passenger counts.
3. Most trips involve 1 or 2 passengers.
4. Higher passenger counts (5–6 passengers) are less frequent.
5. The fare distribution looks similar regardless of how many passengers are in the vehicle.
6. Correlation is weak meaning fare amount is not dependent on number of passengers



Tip Amount vs Trip Distance:

1. There is an upward trend — longer trips generally have higher tip amounts.
2. Most short trips (under 20 miles) have small tips.
3. Some longer trips show higher tips, especially in the mid-distance range.
4. There are many zero-tip trips across all distances.
5. A few high tip values appear in shorter distances, possibly due to generous tipping behaviour
6. Moderate Correlation meaning tip is sometimes dependent on distance and duration but tipping behaviour is not perfectly dependent on distance

3.1.8. Analyse the distribution of different payment types



proportion

payment_type

1	81.49
2	17.27
3	0.49
4	0.75

1. Maximum payments are made through Credit card payments that means Customers prefer digital or cashless transactions.
2. Cash payments are significantly lower than credit card payments.
3. “No Charge” and “Dispute” transactions are very rare.
4. The majority of customers prefer digital payment methods.

3.1.9. Load the taxi zones shape file and display it

Used the geo pandas library to fetch the taxi zone shape file file, we saw columns like:

1. OBJECT ID
2. Shape Leng
3. Shape Area
4. Zone
5. LocationID
6. Borough
7. Geometry

These values clearly represent the geographic boundaries of pickup and drop off zones for spatial analysis.

The LocationID column corresponds to the pickup and drop off location identifiers used in the trip records.

3.1.10. Merge the zone data with trips data

The zones dataset was merged with the trip records using the pickup location identifier. A left join was performed between PULocationID from the trip data and LocationID from the zones data to append zone name and borough and other column information to each trip while retaining all trip record.

Some trips did not have corresponding zone information, resulting in missing values. This could be due to certain pickup locations not being a part of the taxi zone shape file or being outside of NYC taxi zones.

3.1.11. Find the number of trips for each zone/location ID

PULocationID tripCount		
0	1	214
1	2	2
2	3	40
3	4	1861
4	5	13

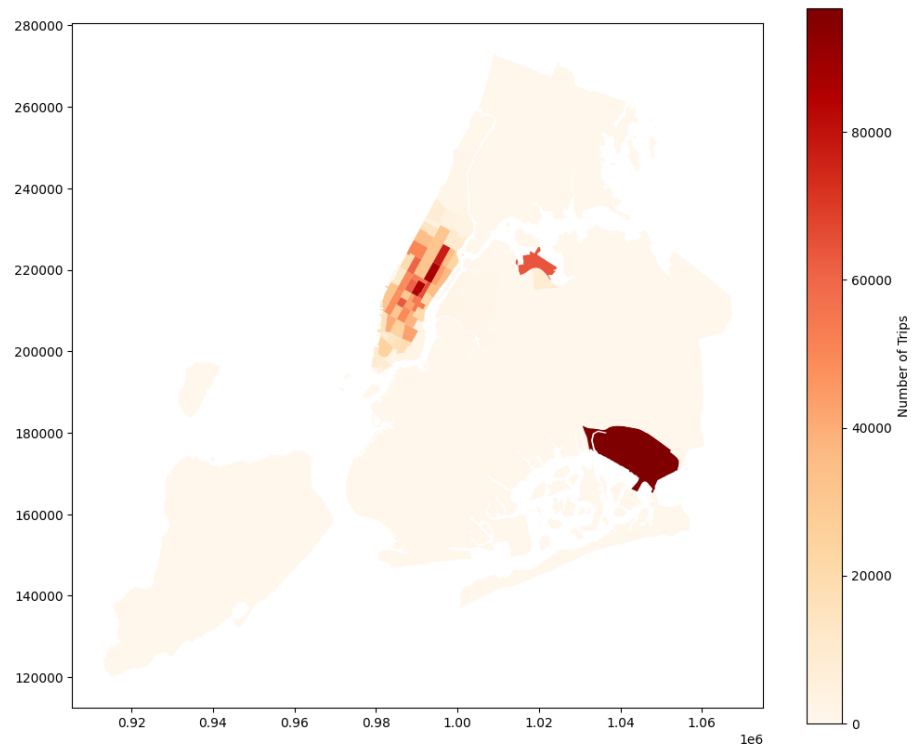
Only showing 5 records in the above image as a representation. The dataset was grouped based on pickup location ID to compute the total number of trips originating from each zone. This helped us in determining pickup areas where there is a high, moderate and low demand areas within the city.

3.1.12.Add the number of trips for each zone to the zones dataframe

The aggregated trip counts were merged with the zones Geo Data frame using the location identifiers. A left join was performed to ensure all zones were retained.

Missing trip counts were replaced with zero and the resulting Geo Data frame was used to visualise demand distribution across taxi zones.

3.1.13.Plot a map of the zones showing number of trips



	LocationID	zone	borough	tripCount
131	132	JFK Airport	Queens	96815
236	237	Upper East Side South	Manhattan	86903
160	161	Midtown Center	Manhattan	85944
235	236	Upper East Side North	Manhattan	77514
161	162	Midtown East	Manhattan	65633

Only showing 5 records in the above image as a representation. A choropleth map was generated using the GeoDataFrame, where zones were colour coded based on total trip counts. Darker shades represent zones with higher taxi activity whereas lighter shades represent fewer taxi trips. The map reveals spatial concentration of taxi demand, particularly in high-density urban areas.

Based on the representation table above we could conclude: Manhattan Dominates Taxi Activity and most high-density zones are concentrated in Manhattan. Midtown Center, Upper East Side South, Upper East Side North, and Midtown East are among the busiest areas.

3.1.14. Conclude with results

The exploratory analysis reveals that NYC taxi demand is strongly influenced by:

- Time of day (commute peaks)
- Day of week (mid-week dominance)
- Seasonality (spring and fall peaks)
- Geographic concentration (Manhattan and airports)
- Distance-based pricing structure
- Moderate behavioural tipping patterns

EDA analysis shows that fare pricing is primarily distance-based, with a very strong correlation between trip distance and fare amount. Trip duration also influences fare but to a lesser extent. Passenger count has almost no impact on total fare.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Routes with the lowest average speeds were identified by calculating the mean distance and mean trip duration for each pickup and dropoff pair at different hours of the day. Low-speed routes can indicate traffic/ congestion or operational inefficiencies. These routes are likely to experience bottlenecks during peak hours and may benefit from route optimisation or demand management strategies.

Only showing 10 records in the below image as a representation. Based on the 10 locations in the image we can determine:

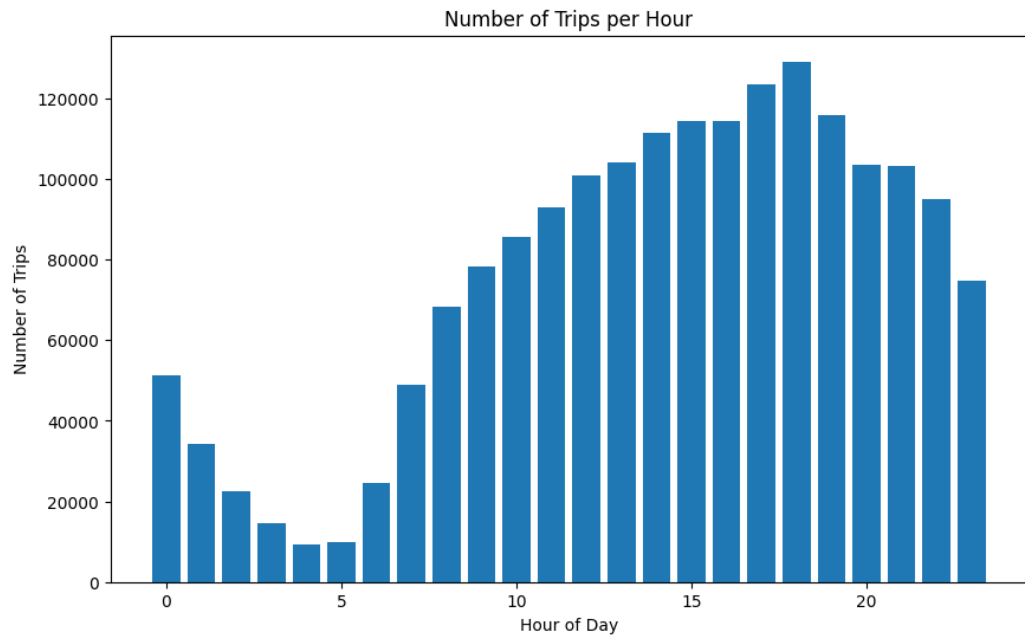
1. Extremely low average speeds are seen on very short trips, where pickup and drop-off locations are the same.

PULocationID	DOLocationID	pickupHr	avg_distance	avg_duration	trip_count	avg_speed_mph
258	258	1	0.02	0.762500	1	0.026230
194	194	16	0.01	0.204444	1	0.048913
45	45	10	0.05	0.840556	1	0.059484
124	129	12	0.10	1.388333	1	0.072029
134	265	15	0.10	1.354444	1	0.073831
67	215	15	0.10	1.280556	1	0.078091
75	203	13	0.10	1.057222	1	0.094587
237	5	15	0.10	1.052222	1	0.095037
89	205	15	0.10	1.041667	1	0.096000
220	236	11	0.10	1.032222	1	0.096878

These trips can represent low speeds due to very small distances combined with short durations.

2. Slow routes occur during afternoon hours (12 PM - 4 PM), indicating higher congestion during peak business activity hours.
3. Routes within high-density Manhattan zones tend to show slower speeds, reflecting traffic bottlenecks in commercial and residential hubs.

3.2.2. Calculate the hourly number of trips and identify the busy hours



1. The busiest hour of the day is 6 PM with 129,184 trips, indicating peak evening demand which could be due to office returns and people leaving for dining.
2. Trip demand gradually increases from early morning 5-6 AM and peaks during the evening between 4PM - 7PM.
3. The lowest demand is observed between 3 AM and 5 AM representing minimal travel during late-night hours.

4. Demand remains relatively high between 12 PM and 9 PM indicating movement during business and social hours.

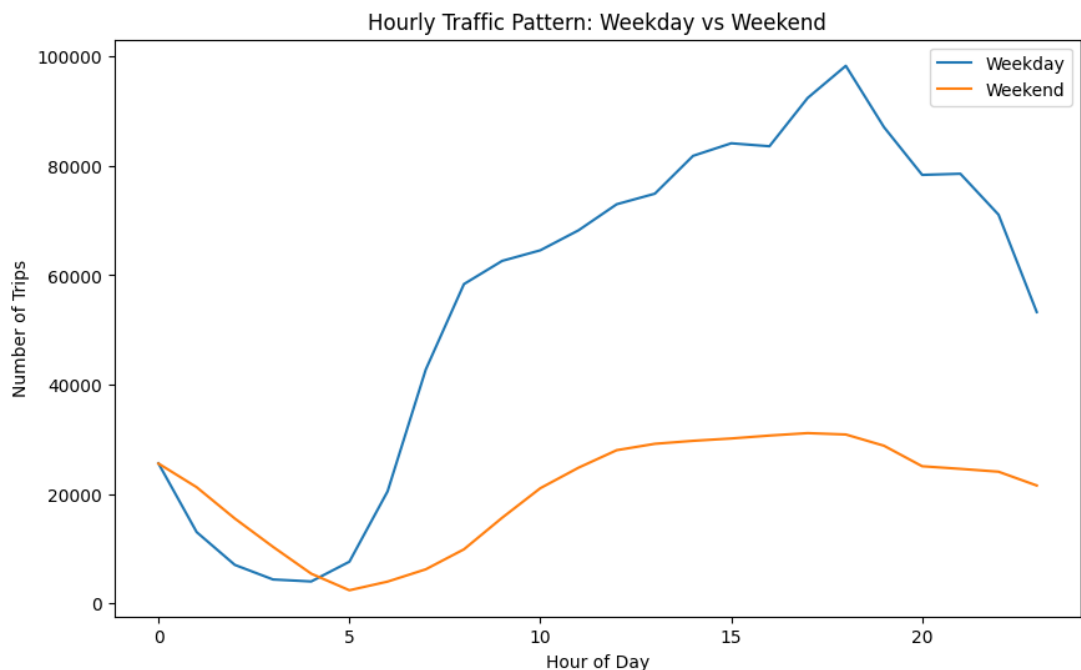
3.2.3. Scale up the number of trips from above to find the actual number of trips

Since the sample dataset represents only a 5% sample of the original data, the actual number of trips was estimated by dividing the sampled trip counts by the sampling fraction (0.05). This scaling provides an approximation of the total trips for the busiest hours.

pickupHr	trip_count	actual_trip_count
18	18	129184
17	17	123557
19	19	115915
15	15	114292
16	16	114281

1. The evening peak window around 3 PM - 7 PM accounts for the highest traffic volumes.
2. The absolute busiest hour is 6 PM, with approximately 2.58 million trips annually.
3. Demand increases sharply post 3 PM, indicating post work travel commutes.

3.2.4. Compare hourly traffic on weekdays and weekends



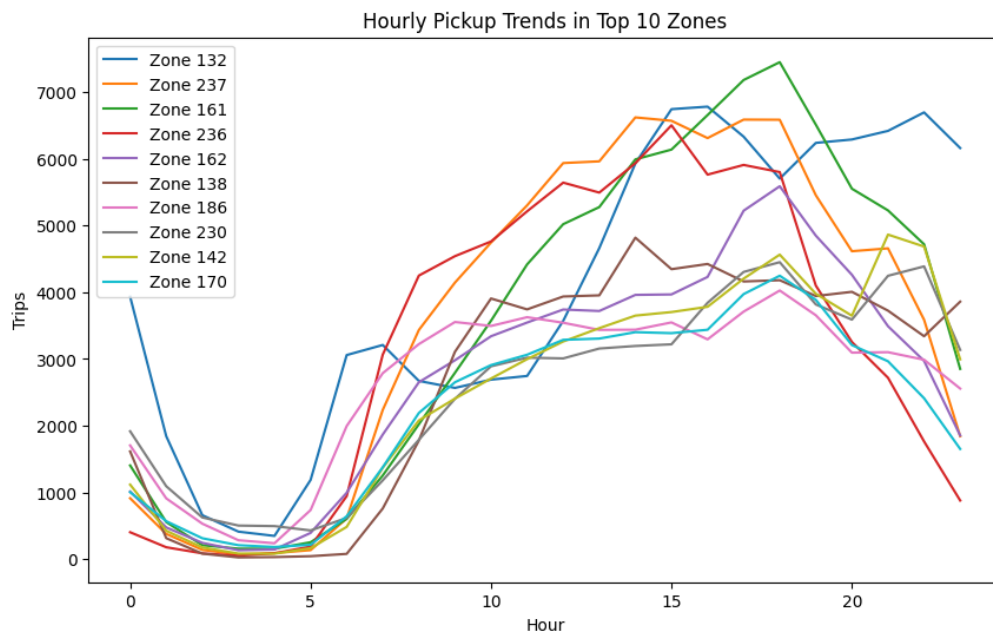
1. Weekday demand is significantly higher than weekend demand
2. On weekdays, traffic rises sharply from 7 AM onwards and peaks at 6 PM, clearly indicating a work and business commuting pattern.
3. Weekends show a delayed rise in demand and peaks around 2 - 6 PM, reflecting leisure and social travel.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

PULocationID	pickup_count
132	96815
237	86903
161	85944
236	77514
162	65633
138	64173
186	63470
230	61314
142	60884
170	54490

Hourly Pickup Zones:

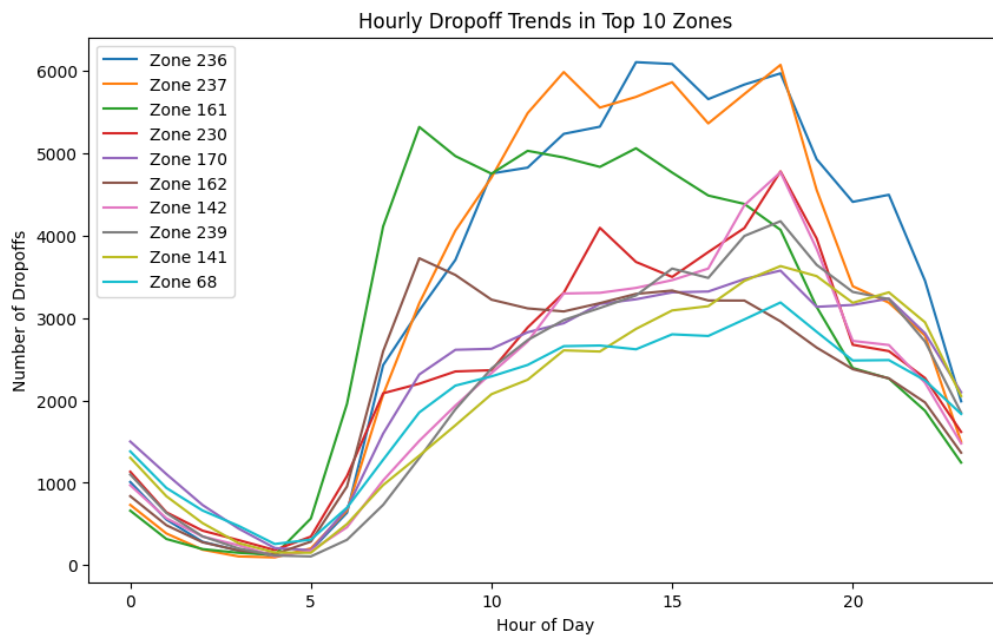
1. 132 - JFK Airport - 96,815 trips
2. 237 - Upper East Side South - 86,903 trips
3. 161 - Midtown Center - 85,944 trips
4. 236 - Upper East Side North - 77,514 trips
5. 162 - Midtown East - 65,633 trips



DOLocationID	dropoff_count
236	81269
237	77556
161	71643
230	56396
170	54311
162	52248
142	51493
239	51258
141	48448
68	46352

Hourly Pickup Zones:

1. 236 - Upper East Side North - 81,269 trips
2. 237 - Upper East Side South - 77,556 trips
3. 161 - Midtown Center - 71,643 trips
4. 230 - Times Square - 56,396 trips
5. 170 - Murray Hill - 54,311 trips



3.2.6. Find the ratio of pickups and dropoffs in each zone

Highest 10 Pickup and drop off ratios:

1. Zone 199 shows an infinite ratio because it has pickups but zero drop-offs. But pickup count is only 2 trips which is extremely low, so this is statistically insignificant and should not be treated as a major pattern.
2. Zone 70 (ratio=8.3) has very high pickup dominance compared to drop-offs. This suggests it may be a residential where passengers frequently start trips.
3. Zone 132 (ratio = 4.62) - JFK - shows strong pickup dominance. This is expected, as many passengers leave the airport via taxi after arrival.
4. Zones 138 and 186 also show strong pickup-heavy patterns, indicating these areas function more as trip origin hubs rather than destinations.

LocationID	pickup_count	dropoff_count	pickup_drop_ratio
199.0	2.0	0.0	inf
70.0	8353.0	1001.0	8.344655
132.0	96815.0	20961.0	4.618816
138.0	64173.0	22249.0	2.884309
186.0	63470.0	40114.0	1.582241
43.0	30751.0	22367.0	1.374838
114.0	24113.0	17539.0	1.374822
249.0	40404.0	30466.0	1.326200
162.0	65633.0	52248.0	1.256182
161.0	85944.0	71643.0	1.199615

Lowest 10 Pickup and drop off ratios:

1. Zones like 1 and 257 show extremely low ratios (0.04–0.03), meaning they receive significantly more drop-offs than pickups. For example:
 - Zone 1: 214 pickups vs 5,318 drop-offs
 - Zone 257: 29 pickups vs 758 drop-offs
2. These zones likely represent either residential neighbourhoods or airport arrival terminals.
3. Strong drop-off ratio may also indicate potential vehicle accumulation, where taxis may cluster after completing trips.

LocationID	pickup_count	dropoff_count	pickup_drop_ratio
176.0	0.0	12.0	0.000000
99.0	0.0	3.0	0.000000
30.0	0.0	18.0	0.000000
245.0	0.0	30.0	0.000000
27.0	1.0	39.0	0.025641
221.0	1.0	34.0	0.029412
257.0	29.0	758.0	0.038259
1.0	214.0	5318.0	0.040241
115.0	1.0	23.0	0.043478
198.0	52.0	990.0	0.052525

3.2.7. Identify the top zones with high traffic during night hours

PULocationID	pickup_count	DOLocationID	dropoff_count
79	15550	79	8313
132	14544	48	6874
249	12472	170	6265
48	10463	68	5858
148	9629	107	5756
114	8749	141	5272
230	8206	263	4977
186	6968	249	4944
164	6143	230	4641
68	6050	148	4382

1. Zone 79 and 132 are the busiest night pickup zones.
2. Likely high-activity nightlife or transport hubs.
3. Zone 79 has both pickup and dropoff at night showing high two-way traffic.
4. Zones 48, 249, 230, 148, 68 appear in both pickup and dropoff and show consistent movement at night in these zones .
5. Zone 170 appears strongly in dropoffs, possibly a residential destination zone at night. People coming to home after office hours.

3.2.8. Find the revenue share for nighttime and daytime hours

Revenue share for night time: 12.06%

Revenue share for night time: 87.94%

Observations:

1. Nighttime trips contribute only 12.06% of total revenue.
2. The majority of revenue (87.94%) is generated during daytime hours.
3. This shows that taxi demand and earning potential are significantly higher during the day.

3.2.9. For the different passenger counts, find the average fare per mile per passenger

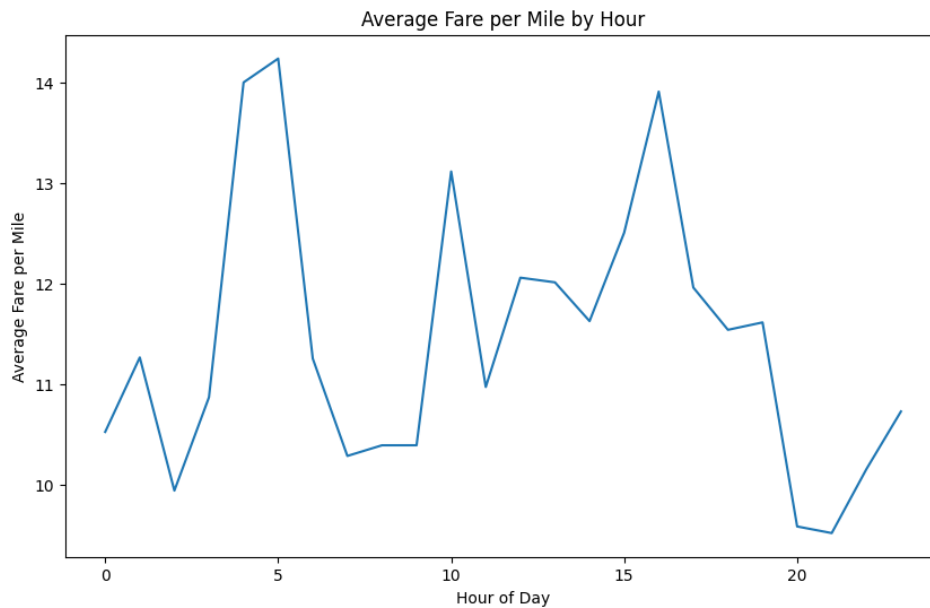
passenger_count	avg_fare_per_mile	fare_per_mile_per_passenger
1.0	11.012470	11.012470
2.0	12.864801	6.432401
3.0	11.724296	3.908099
4.0	17.452908	4.363227
5.0	8.548072	1.709614
6.0	8.104462	1.350744

1. Single passenger trips - Average fare per mile = \$11.01
2. Two passengers - Average fare per mile = \$12.86 so Fare per mile per /3 passenger = \$6.43
3. Three passengers - Average fare per mile = \$11.72 so Fare per passenger = \$3.91

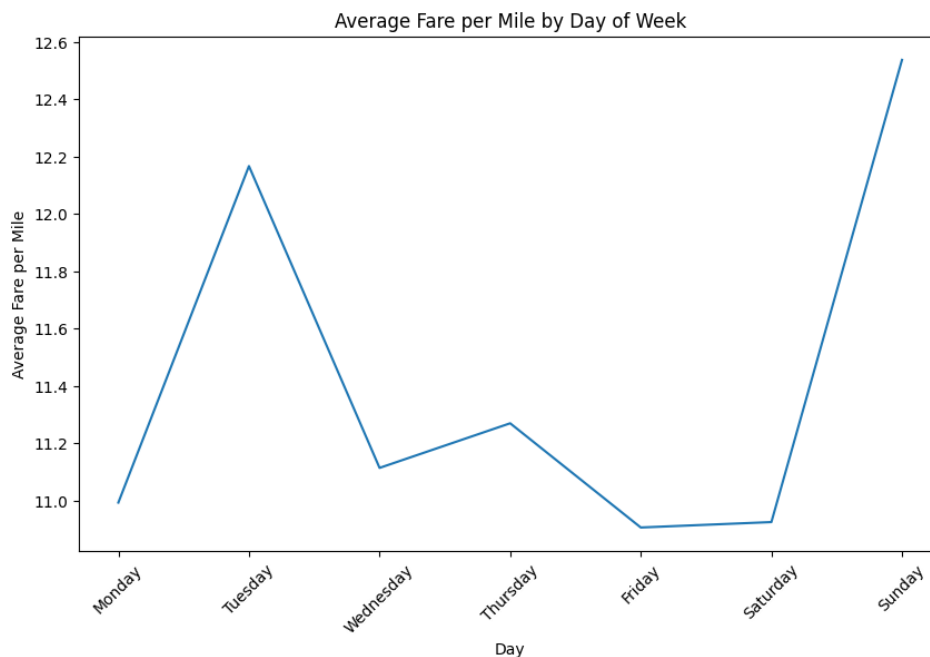
Observations:

1. As passenger count increases, cost per passenger decreases significantly.
2. The NYC taxi ride system is economically favourable for shared rides.
3. Solo riders pay the highest effective per-person cost.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

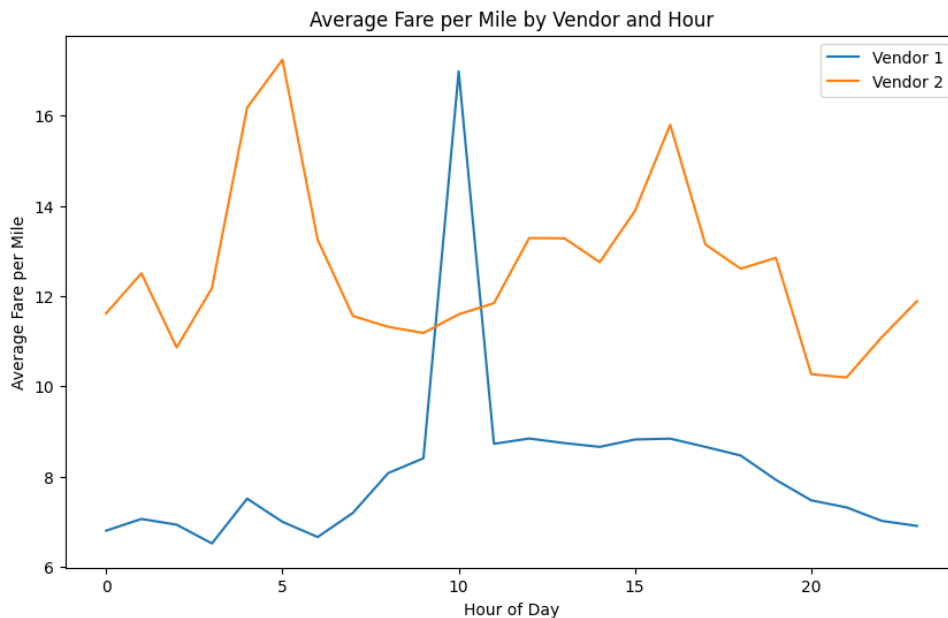


1. The average fare per mile fluctuates between \$9.5 and \$14.5.
2. Early morning hours 4-6 AM show relatively higher fare per mile.
3. Another noticeable increase occurs during late afternoon (around 4-6 PM).
4. The lowest fare per mile appears during late evening (around 8-9 PM).
5. **Higher fare per mile during early morning may be due to:** Lower trip distances and Airport trips
6. **Late afternoon spikes may be due to:** Peak demand periods
Congestion-related pricing component



1. Sunday has the highest average fare per mile around \$12.5?
2. Friday and Saturday show slightly lower fare per mile around \$11
3. Higher Sunday fare per mile may indicate: More airport trips, More leisure travel or Shorter, high-value trips
4. Slightly lower Friday/Saturday fare per mile may reflect: Longer rides or More shared or group travel

3.2.11. Analyse the average fare per mile for the different vendors



1. Vendor 2 consistently charges a higher average fare per mile than Vendor 1 across most hours of the day.
2. Vendor 2 shows strong peaks during early morning around 4-6 AM and late afternoon around 4 PM.
3. Vendor 1 shows relatively stable pricing except for a noticeable spike around 10 AM which could be a surged pricing.
4. Vendor 2 appears to follow a more aggressive or premium pricing strategy.
5. Vendor 1 seems to maintain more stable and lower pricing.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

VendorID	distance_tier	avg_fare_per_mile
1	2 to 5 miles	6.381262
1	More than 5 miles	4.425249
1	Up to 2 miles	10.673070
2	2 to 5 miles	6.549258
2	More than 5 miles	4.503240
2	Up to 2 miles	17.929401

Short Trips (Up to 2 miles)

Vendor 1: \$10.67 per mile

Vendor 2: \$17.93 per mile

Vendor 2 charges significantly higher for short trips.

Medium Trips (2 to 5 miles)

Vendor 1: \$6.38 per mile

Vendor 2: \$6.55 per mile

Both vendor have almost a similar pricing

Long Trips (More than 5 miles)

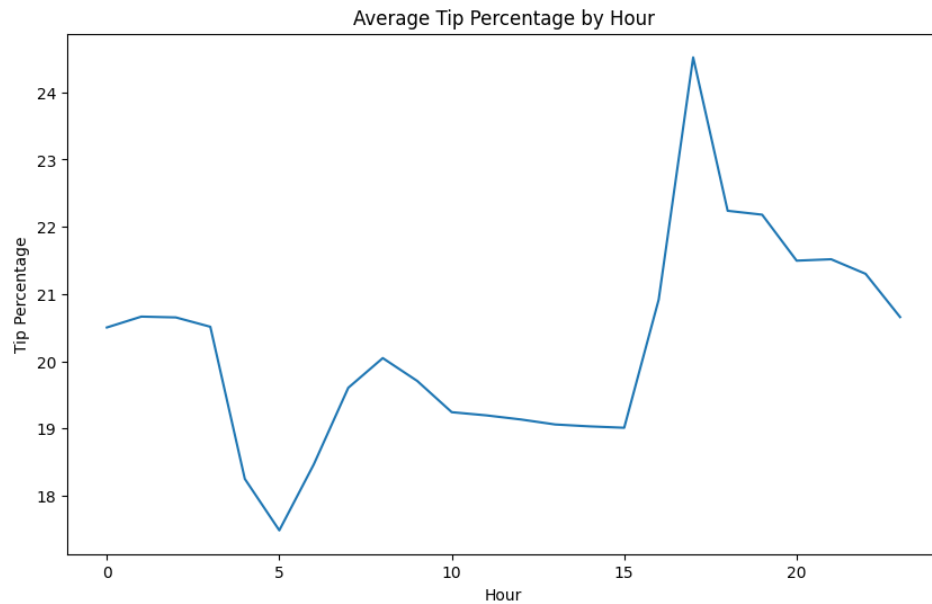
Vendor 1: \$4.43 per mile

Vendor 2: \$4.50 per mile

Both vendor have almost a similar pricing

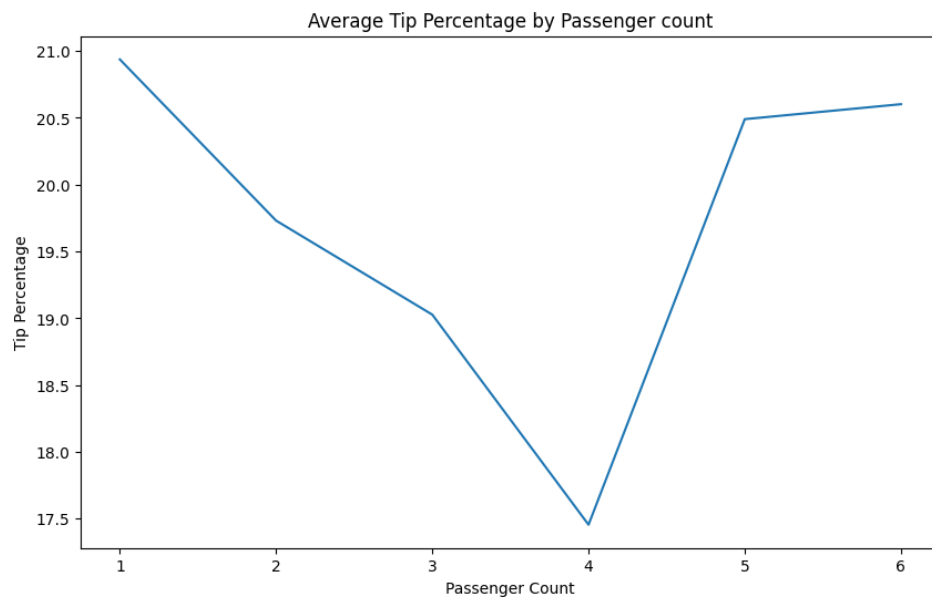
3.2.13. Analyse the tip percentages

Tip percentage by hour

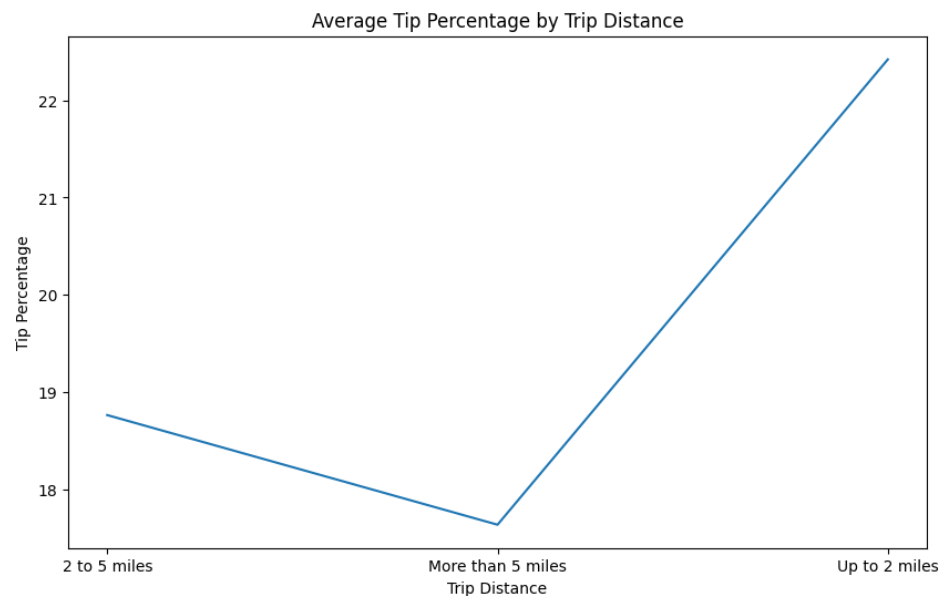


1. Tip percentages remain relatively stable throughout most of the day, generally between 19% and 22%.
2. The lowest tipping occurs during early morning hours around 4-6 AM, where the average tip drops below 18%.
3. Tip percentages gradually increase during the day.
4. The highest tipping occurs during the evening hours around 5-7 PM, peaking at approx. 25%.

Tip percentage by passenger count

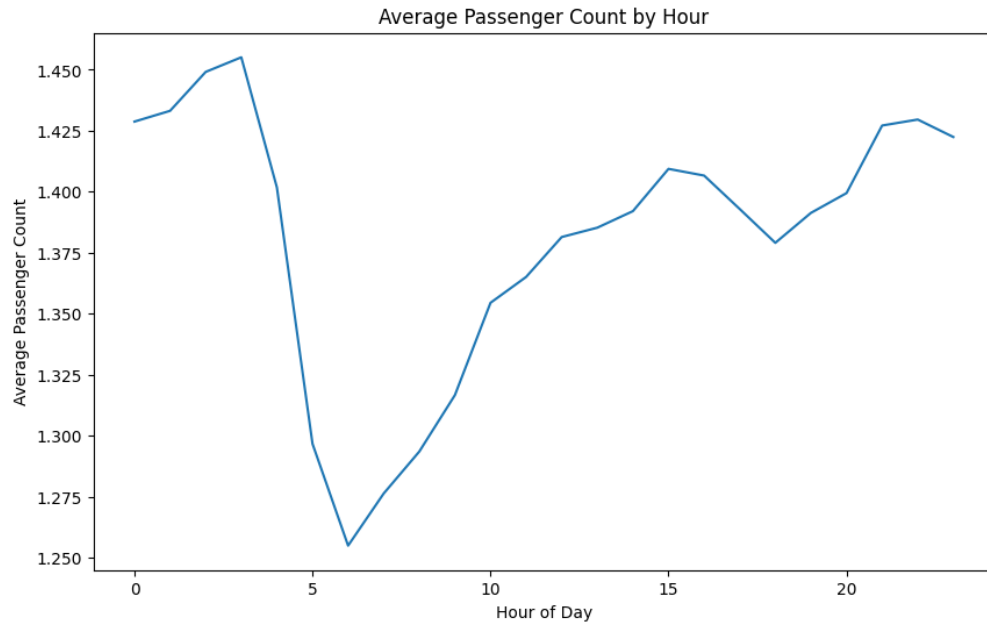


1. Trips with 1 passenger have the highest average tip percentage (around 21%).
2. As passenger count increases from 1 to 4, the tip percentage gradually decreases.
3. Trips with 4 passengers show the lowest tipping percentage (around 17-18%).
4. For 5 and 6 passengers, the tip percentage increases again, reaching around 20-21%.

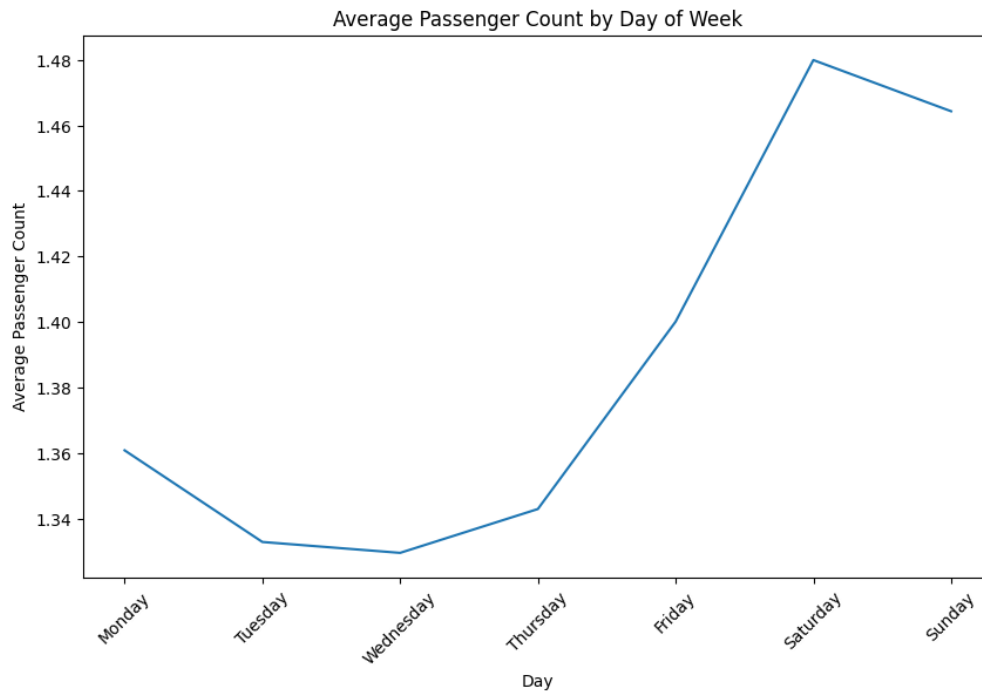


1. Tip percentage is highest when the trip distance is under 2 miles
 2. Tip percentage drops to less than 18 percent when trip is more than 5 miles
 3. Tip percentage ranges between 18-19 percent between 2-5 miles.
- This tells us that tip amount is moderately affected by the trip distance

3.2.14. Analyse the trends in passenger count



1. The average passenger count remains relatively stable throughout the day, generally having 1 passenger per trip.
2. The highest average passenger count occurs during late night and early morning hours between 1-3 AM.
3. The lowest average passenger count occurs during early morning commute hours around 5-7 AM.
4. Passenger count gradually increases again during the afternoon and evening.
4. Evening hours between 8-10 PM show slightly higher average passenger counts compared to morning commute hours.

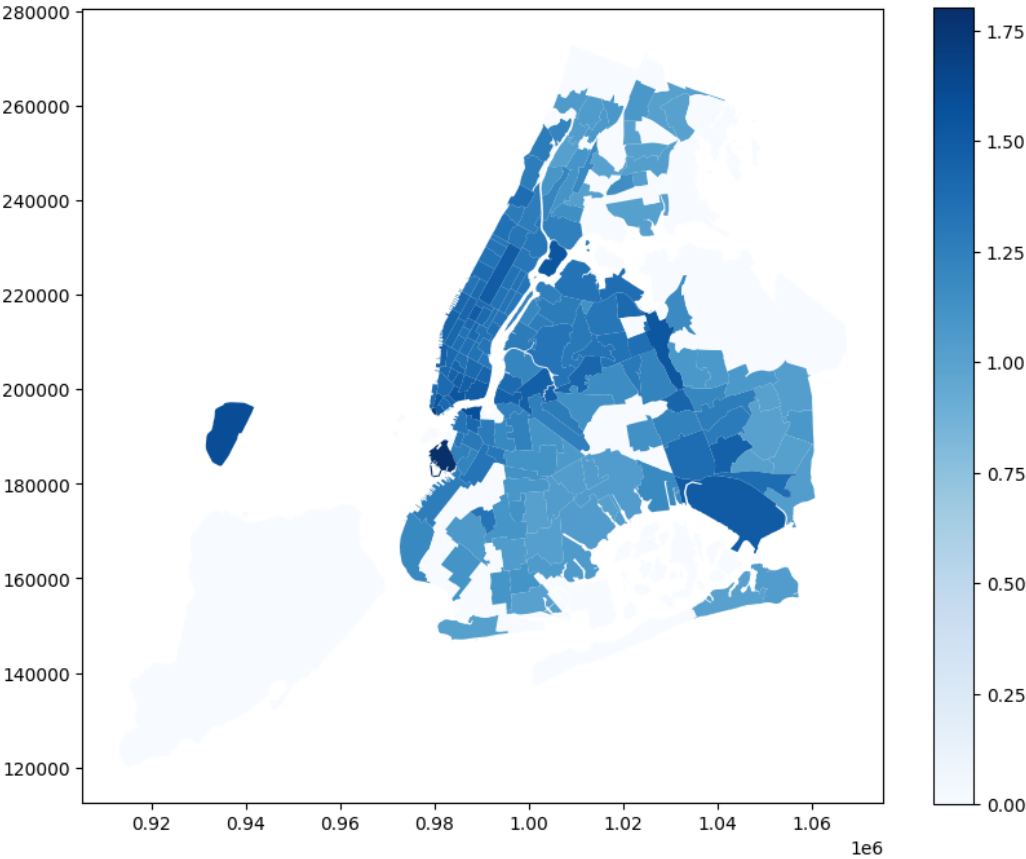


1. The average passenger count is lowest during mid-week (Tuesday and Wednesday)
2. Passenger count begins to increase from Thursday onward.
3. Friday shows a noticeable rise in average passengers.
4. Saturday has the highest average passenger count.
5. Sunday remains high but slightly lower than Saturday

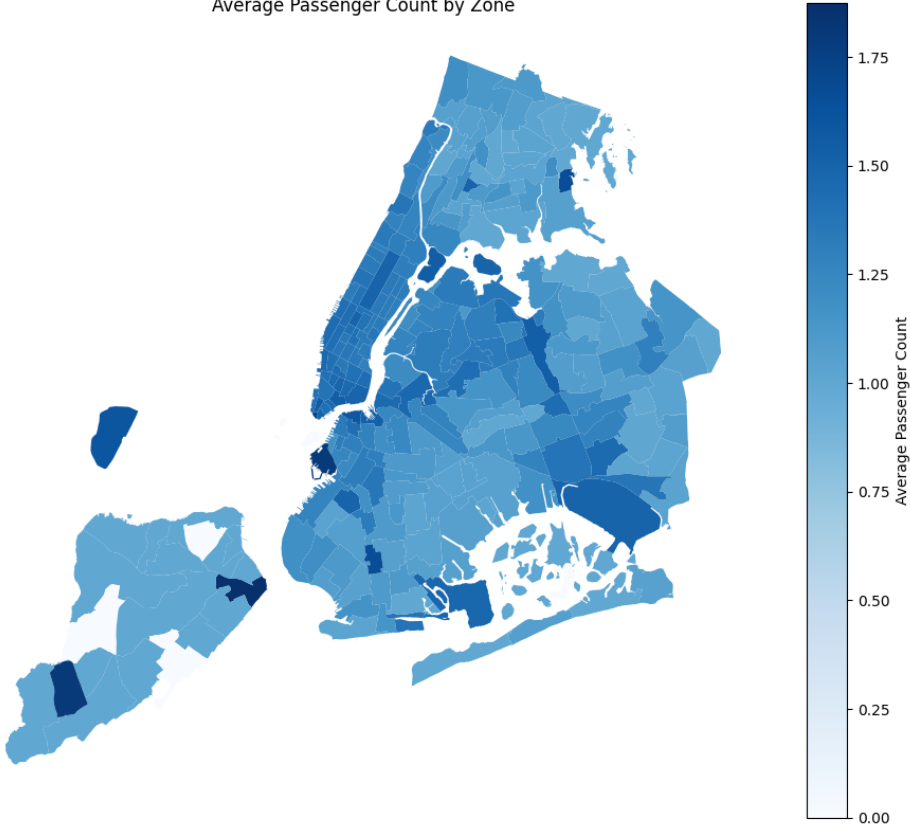
Observations:

1. Weekday trips (especially Tuesday and Wednesday) are mostly solo travellers.
2. Weekend trips involve more group travel.

3.2.15. Analyse the variation of passenger counts across zones



Average Passenger Count by Zone

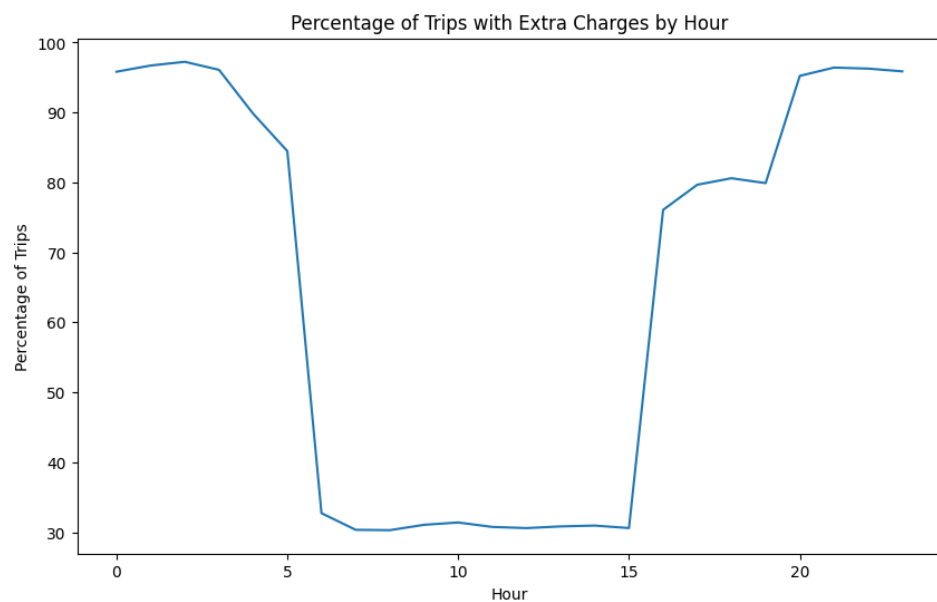


The choropleth maps display the average passenger count per trip across different taxi zones in New York City.

1. Darker blue zones indicate higher average passenger counts.
2. Lighter zones indicate lower average passenger counts.

Observation: The overall average passenger count across zones stays mostly between 1-2 passengers per trip, meaning most taxi rides are still primarily individual or small group travel.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.



Observations:

1. Very High Extra Charges During Late Night between 12 AM - 4 AM:

This could indicate reasons like: Night surcharges, Reduced driver availability or higher cost during late hours

2. Drop Between 6 AM - 3 PM:

This period represents standard working hours and regular pricing without any or heavy surcharge applied on the rides during business hours

3. Increase After 4 PM: Between 4 PM – 8 PM, surcharges increase sharply

This could be due to the following reasons: Evening rush hour congestion surcharge and Increased demand during return from offices and business related.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

1. Use congestion-aware routing during peak hours to avoid slow routes.
2. Increase fleet availability during weekday commute peaks (7-9 AM, 4-7 PM).
3. Increase fleet allocation during weekend late-night hours.
4. Reposition idle vehicles toward zones with high pickup/dropoff imbalance.
5. Maintain dedicated fleet presence in airport zones.
6. Use historical hourly trends to predict demand spikes and pre-dispatch vehicles.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

1. Business District Strategy:
 - Position more cabs in commercial hubs during weekday mornings.
 - Shift fleet toward residential zones in the evening.
2. Weekend Strategy:
 - Allocate more cabs to entertainment and nightlife zones.
 - Increase availability between 10 PM-2 AM on weekends.
3. Airport Strategy:
 - Maintain steady supply near airport zones.
 - Adjust fleet allocation based on arrival/departure schedules.
4. Low-Demand Zones:
 - Avoid oversupply in zones with consistently low trip counts.
 - Use dynamic rebalancing instead of static allocation.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

1. **Distance-Based Tier Pricing:**
 - Since short trips have higher fare per mile, review the base fare structure for ultra-short trips.
 - Offer incentives or discounts for longer-distance trips.
2. **Night-Time Pricing.**
 - Apply mild dynamic pricing during high-demand night hours.
 - Offer driver incentives instead of steep passenger surcharges.

- Congestion-Based Pricing.
 - Introduce congestion-adjusted pricing during peak slow hours.
 - Consider transparent route-based fare breakdown.
3. **Payment & Tipping Strategy:**
- Encourage digital payments to increase recorded tip percentages.
 - Provide preset tip suggestions to improve driver earnings.
 - Offer loyalty incentives for high-frequency riders.
4. **Vendor Competitiveness:**
- Monitor fare per mile differences across vendors.
 - Adjust pricing in high-competition zones to retain customers.

Final Strategic Conclusion:

1. Taxi demand is strongly influenced by:

- Time of day (commute vs leisure)
- Day of week (weekday vs weekend)
- Zone type (business, residential, airport)
- Trip distance
- Passenger composition
- Payment behaviour

2. An optimised strategy should:

- Dynamically allocate fleet based on temporal demand.
- Rebalance vehicles using pickup/dropoff ratios.
- Implement tiered and time-aware pricing strategies.
- Improve routing efficiency during congestion.
- Leverage data insights for continuous demand forecasting.

3. This integrated approach will maximize revenue, improve operational efficiency, and enhance customer experience while maintaining competitiveness.